MILITARY SPECIFICATION

PROPELLANT M9

FOR USE IN CHARGE, PROPELLING, M36A2

1. SCOPE

1.1 This specification covers M9 Propellant for use in the following Cartridge employed with the 4.2 Inch Mortar, M30: Cartridge, HE, M329A2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-652 - Propellants Solid for Cannons Requirements and Packing
MIL-STD-1168- Lot Numbering of Ammunition
MIL-STD-1172- Acceptance and Description Sheets for Propellants and Explosives

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for

DRAWINGS

ARMY

9244181 - Bag, Loading Assembly
9235654 - Cartridge, HE, M329A2

FSC: 1375

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2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

Title 49, Transportation: Parts 171 - 179

AMC PAMPHLET 706-114

Experimental Statistics, Section 5, Tables

The Code of Federal Regulations is available from the Superintendent of Documents, Government Printing Office, Washington, D.C., 20402. Orders for the above publication should cite: 49 CFR 171-179 (latest revision).

3. REQUIREMENTS

3.1 Material.-The propellant shall be M9 flake propellant with an advisory diameter of 0.059 inches and a nominal thickness of 0.006 inches and shall conform to the requirements of MIL-STD-652.

3.2 Proving Ground.-The propellant, when fired in the 4.2 inch mortar, M30 and with the weight of projectile (21.99 lbs) for which it was manufactured, shall comply with the following requirements when tested as specified in 4.3.

3.2.1 Assessment.-At 70 degrees Fahrenheit (°F) and standard weapon conditions, the propellant shall be capable of charge assessment to a velocity of 313 feet per second at charge 0 without an associated expected pressure greater than 2800 psi or less than 1800 psi.

3.2.2 Uniformity of velocity.-The standard deviation in velocity for the lot at 70°F shall not exceed 5.0 feet per second.

3.3 Graphite (Glaze).-The graphite content shall not exceed 0.40 percent when determined as specified in 4.3.5.

3.4 Workmanship.-The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. No reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification unless otherwise specified.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescirbed requirements. Reference shall be made to Standard MIL-STD-109, in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information acccompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of batches of product inspected (shall be made available upon request by the contracting officer).
c. Results obtained, by defect code, for all inspections performed (shall be made available upon request by the contracting officer).
d. Drawing, specification number, and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contracting documents (shall be made available upon request by the contracting officer).
f. Number of items in the lot.
g. Date submitted.
h. Eight copies of propellant description sheet SMU Form 1060 containing the applicable data. (see 6.8).

4.1.2 The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.2 Inspection provisions

4.2.1 Lot formation.-The term "lot as used throughout this specification, refers to an inspection lot, which is defined as an essentially homogeneous quantity of propellant from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. A lot shall consist of one or more sub-lots or one or more batches of M9 propellant produced by one manufacturer, in accordance with the same specification, and same specification revision under one continuous set of operating conditions. Each lot shall consist of that quantity of M9 propellant which has been subjected to the same unit chemical or physical process intended to make the final product homogeneous.

4.2.2 Testing

4.2.2.1 Materials (see 3.1).-Each lot of propellant shall be sampled and tested for composition, form, dimensions and chemical-physical properties in accordance with MIL-STD-652.

4.2.2.2 Proving ground (see 3.2).-Each lot of propellant shall be sampled for proving ground tests in accordance with MIL-STD-652.

4.2.2.2.1 Assessment (see 3.2.1).-Major defect, Code No. 01001 (see 6.4).-Failure to comply with the requirement stated in 3.2.1 as determined in MIL-STD-414 section B part II (AQL of 0.40) shall cause rejection of the lot.

4.2.2.2.2 Velocity Uniformity.-(70°F) (See 3.2.2 as applicable) Major defect, Code No. 02001.-If the product of the factor (see 6.5) times the velocity standard deviation of the rounds tested in accordance with 4.3.3 exceeds the applicable requirement the lot shall be rejected. For initial production lots a ten round sample shall be tested. (see 6.6)

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4.3 Test methods and procedures

4.3.1 Proving ground tests.-These tests shall be conducted at a Government proving ground (designated by the contracting officer) in accordance with the applicable Proving Ground Acceptance Test Procedure.

4.3.2 Charge assessment

4.3.2.1 Charge Slope Method.-Three or more groups of rounds loaded with selected charge weights of propellant (weights selected should bracket expected service charge weight) shall be conditioned and fired at 70°F. From the data obtained, a charge weight shall be calculated which will yield the prescribed service velocity. This calculated charge weight shall then be loaded into rounds for the uniformity series (see 4.3.3). The difference between the corrected velocity obtained during the uniformity series and the service velocity shall be compensated for by a final adjustment in charge weight using the slope of the vel-chg weight curve previously established.

4.3.2.2 Constant Slope Method.-After ten (10) lots of a given production series have been fired to provide a good knowledge of the propellant characteristics, this charge slope method of assessment may be discontinued and the charge weight determined by the constant-slope method. A composite velocity vs charge weight vs pressure curve can be established from test results generated. And this curve can then be used as in 4.3.2.1 above to calculate the charge weight needed to achieve service velocity . This calculated charge weight is then to be loaded into rounds for the uniformity series and final adjustment is to be made as indicated above for the charge slope method.

4.3.3 Uniformity series

4.3.3.1 Initial production

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4.3.3.1.1 70°F.-Ten rounds of M329A2 Cartridges shall be assembled with the assessed charge of propellant, conditioned to 70°F (as indicated by thermocoupled control rd) and fired alternately with calibration rounds. The velocity and pressure means, as well as the standard deviations associated with these parameters shall be calculated. All results shall be corrected to standard shell weight (21.99 lbs) and standard weapon/firing conditions (i.e. the difference between calibration performance under test conditions and standard conditions) and to recommended charge weight conditions.

4.3.3.2 Low Temperature Firings.-For the first ten lots of propellant produced by a manufacturer in accordance with this document, a low temperature uniformity series shall be conducted in addition to the 70°F firings. This series shall consist of 10 rounds assembled in the same manner as 4.3.3.1, but conditioned at a temperature of -50°F prior to firing. Velocity and pressure readings shall be taken and reported for information only.

4.3.3.3 Subsequent Production.-Seven rounds of Cartridges shall be assembled and tested at 70°F. The velocity and pressure means as well as the standard deviations associated with these parameters shall be calculated. All results at 70°F shall be corrected to standard shell weight (21.99 lbs) and standard weapon/firing conditions and recommended charge weight conditions.

4.3.4 Refiring.-If for any reason the proving ground considers that test conditions have detrimentally affected the test result, additional charges as required shall be tested.

4.3.5 Graphite (glaze) determination.-The graphite content shall be determined in accordance with MIL-STD-652.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be in accordance with Standard MIL-STD-652.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Packing required (see 5.1)
c. Process.-Details of the manufacturing process and the equipment used by the Contractor will be submitted to the procuring activity in writing prior to commencement of manufacture.

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6.2 Standard deviation.-The standard deviation shall be calculated with (n-1) as the devisor in a standard statistical technique, equivalent to that shown in Standard MIL-STD-414, Section B, Example B-1.

6.3 Submission of test data.-In addition to the normal distribution of records, when the propellant is procured by the Department of the Army, one copy of all ballistic test data shall be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-P, and one copy to ATTN: SMUPA-ADEP-2, Dover, New Jersey 07801.

6.4 Inspection code number.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.5 Factors.-The factors, as taken from Table A-21 of AMC Pamphlet 706-114 using the 95 per cent upper confidence limit (A.05) and n-1 degrees of freedom, make allowances for the probability that a standard deviation from a smaller sample size exceeds the true standard deviation by chance alone.

6.6 Initial production lots.-The initial production lots represent the first three consecutive lots from each manufacturer which have been produced and met the acceptance criteria of this specification. Lapse in production of 90 days or more shall cause testing to revert back to initial production plan.

6.7 Tolerance.-A tolerance of minus 5 degrees F. applies to all -50 and 145 degrees F. requirements and a tolerance of plus or minus 2 degrees F. to all 70 degrees F. requirements.

6.8 Submission of SMU Form 1060.-Information contained in propellant description sheet, SMU Form 1060, need not be resubmitted in the form of a certificate provided the provisions regarding certifying of responsible agents in adhered to (see 4.1.2).

6.9 Surveillance sample.-Samples shall be forwarded to Picatinny Arsenal, Dover, N.J., ATTN: SMUPA-FR-G, for the propellant surveillance test program in accordance with MIL-STD-652.

6.10 Test cartridges shall be assembled in accordance with the following drawings:

9244181 - Bag Loading Assembly
9235654 - Cartridge, HE, M329A2

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6.11 Proving ground test summary

Initial Production

Temp (°F)	Sample Size	Vel(fps) (assessed)	Lot Std. Dev. of Vel. (fps)	Expected Pres. (psi)
-50*	10	na	na	na
70	10	313	5.0	1800 - 2800

Subsequent Production

70	7	313	5.0	1800 - 2800

*Information only - Discontinued after 10 lots of a producer have been tested.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project No. 1375-A-005

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